

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGYSixth Semester of B. Tech. (IT) Examination
May 2022**IT354 MACHINE LEARNING**

Date: 05/05/2022, Thursday

Time: 10:00 a.m. To 01:00 p.m.

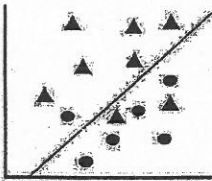
Maximum Marks: 70

Instructions:

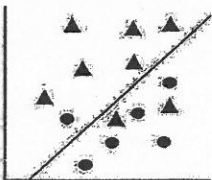
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the following questions. [10]**

- (a) What are some unsupervised learning tasks? [01]
- (b) Why data and computing are considered as challenges for machine learning (ML)? [01]
- (c) How do we quantify goodness of a machine learning model? [01]
- (d) Write your interpretation of following situation in machine learning in one word. [01]
"Training accuracy of algorithm is very high but testing/validation accuracy is very low/not acceptable"
- (e) How does reinforcement learning work? [01]
- (f) "Non-parametric machine learning methods are best suitable for high amount of data". True/False? [01]
- (g) Why do we need validation dataset along with training dataset and testing dataset? [01]
- (h) What is the difference between supervised and unsupervised learning? [01]
- (i) What is precision in following system if want to classify object is triangle or not? [01]



- (j) What is recall in following system if want to classify object is triangle or not? [01]



Q - 2 Answer the following questions. [Any Two] [10]

(a) Explain use of following terms in context of logistic Regression with figure. [05]

i. Decision Boundary

ii. Sigmoid function

iii. Log loss Function

(b) Write a short note on Hierarchical Clustering. [05]

(c) Explain gradient descent algorithm. [05]

Q - 3 Answer the following questions. [Any Three] [15]

(a) Use Linear Regression to calculate the value of dependent variable for the following data when the value of independent variable is 5. [05]

X	Y
20	200
25	275
30	100
10	300
35	150
5	?

(b) Explain Association Rule Learning with an example. [05]

(c) Use k-NN (k=3) to find the class of the data point label marked with "?" in the following table. Use Manhattan distance (absolute distance) instead of Euclidean distance (root of sum of squares): [05]

X1	X2	X3	Y
3	4	1	0
1	4	2	0
2	3	3	0
6	-1	1	1
4	1	-2	1
5	-3	3	?

[P.T.O.]

- (d) There were 10000 chips manufactured in a plant out of which 9500 were normal but 500 were defective. A machine learning algorithm correctly predicts only 9300 of the normal chips as normal and only 450 of the defective chips as defective. Calculate the following with explanation. [05]
- Confusion matrix
 - Accuracy
 - Precision
 - Recall

SECTION – II

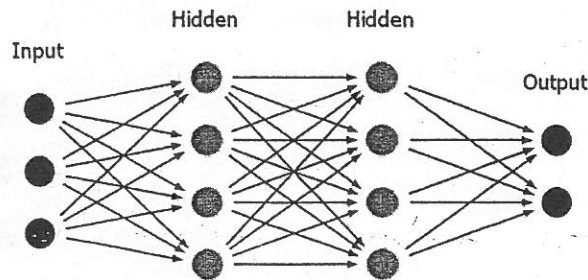
- Q - 4 Answer the following questions. [10]**
- What is meaning of support vectors in SVM? [01]
 - "SVM optimization focuses only on maximizing distance between support vectors" True/False? [01]
 - How perceptron works? Explain with diagram. [01]
 - How to differentiate between Neural Network and Deep Neural Network? [01]
 - Which are suitable applications of CNN? [01]
 - Why pooling layer in CNN are not having any weights? [01]
 - What is projection error in PCA? [01]
 - "PCA is used for feature selection". True/False. [01]
 - "K-means is unsupervised algorithm used to classify data in predefined labels". True/False [01]
 - If I am using all features of my dataset and I achieve 60% accuracy on my training set, but 55% on the validation set, what should I look out for? Considering minimum accuracy as 80%. [01]
- Q - 5 Answer the following questions. [Any Two] [10]**
- Explain back propagation in neural networks with example. [05]
 - Explain convolution and max pooling in convolutional neural networks. Give example. [05]
 - What steps can be taken to mitigate situation of over fitting and under fitting in machine learning model? [05]

[P.T.O.]

Q - 6 Answer the following questions. [Any Three]

[15]

- (a) What is difference between parameters and hyperparameters in a neural network? For the neural network diagram given below, calculate the total number of parameters. Show your reasoning behind the calculations. [05]



- (b) Apply k-means clustering on the following data for K (number of clusters) = 3. [05]

Student Height (cm)	Student Weight (kg)
174	108
165	65
155	61
183	85
140	50
170	72
191	78
145	68

- (c) What is kernel trick in SVM? Explain with example. [05]
- (d) Write a case study on how machine learning can be used in the healthcare industry. The case study must be from the perspective of machine learning life cycle. Consider a specific dataset size, dataset type and application for the case study. [05]